Kuan-Hsun Tu

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EDUCATION

National Taiwan University Sep 2021 - Jun 2026
B.S. in Computer Science and Information Engineering GPA: 3.71/4.30
Taipei, Taiwan

RESEARCH EXPERIENCE

Interactive Perception and Robot Learning Lab, Stanford University
Research Collaborator (Advisor: Prof. Jeannette Bohg) Aug 2026 - Present

- **Reward-Promptable Dexterous Manipulation.** Enabled frozen robot policies to solve novel reward-defined manipulation tasks without retraining through *Policy Futures*.

Embodied AI Lab, National Taiwan University
Research Assistant (Advisor: Prof. Tsung-Wei Ke) Sep 2024 - Present

- **Dexterous Manipulation.** Enabled learning of bimanual manipulation from human demonstrations and generated videos in *DexMan*.
- **Unsupervised Zero-Shot RL.** Improved how Behavioral Foundation Models use novel reward functions to select pretrained behaviors, achieving stronger zero-shot performance in *BLS*.
- **Flow Matching.** Improved generation quality through flow reversal, constructing cluster-wise source distributions in *COT-FM* and learning condition-dependent source distributions in *CNP-Flow*.

PUBLICATIONS AND PREPRINTS

[*] Equal contribution. [†] Corresponding author.

1. **Kuan-Hsun Tu**[*], Chien-Sheng Chiang[†*], Hsin-Wei Chen, Ping-Chun Hsieh, Tsung-Wei Ke
   Task Inference Beyond Least Squares in Behavioral Foundation Models
   NeurIPS 2026 PTA Workshop.
   arXiv · Project Website

2. **Kuan-Hsun Tu**[*], Hsuan-Chi Liu[†*], Jia-Wei Liao[*], Chien-Sheng Chiang, Cheng-Fu Chou, Tsung-Wei Ke
   Learning Conditional Source Distribution via Flow Reversal for Temporal Flow Matching
   arXiv preprint, 2026.
   arXiv · Project Website

3. Chien-Sheng Chiang[*], **Kuan-Hsun Tu**[†*], Jia-Wei Liao[*], Cheng-Fu Chou, Tsung-Wei Ke
   COT-FM: Cluster-wise Optimal Transport Flow Matching
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
   arXiv · Code · Project Website

4. Yash Jangir, Yidi Zhang[*], Kashu Yamazaki[*], Chenyu Zhang[*], **Kuan-Hsun Tu**, Tsung-Wei Ke, Lei Ke, Yonatan Bisk, Katerina Fragkiadaki
   RobotArena ∞: Scalable Robot Benchmarking via Real-to-Sim Translation
   International Conference on Learning Representations (**ICLR**), 2026.
   arXiv · Code · Project Website

5. Jhen Hsieh, **Kuan-Hsun Tu**, Kuo-Han Hung, Tsung-Wei Ke
   DexMan: Learning Bimanual Dexterous Manipulation from Human and Generated Videos
   arXiv preprint, 2025.
   arXiv · Code · Project Website

HONORS AND AWARDS

Student Representative Speaker · NTU Opening Ceremony Sep 2026

1st Place (Coach) · HiPAC 2026, NCHC Performance Optimization Competition Aug 2026

Overall Winner (1st Place) · ACM/IEEE SC25 Student Cluster Competition Nov 2025

2nd Place & Best AI Performance Award · 8th APAC HPC-AI Competition Nov 2025

WORK EXPERIENCE

NTU COOL
Jul 2026 - Present

Backend and Infrastructure Developer

- Developed and operated production backend infrastructure for NTU COOL AI Companion, including FastAPI services, PostgreSQL, GPU-based LLM inference, and monitoring.
- Designed and deployed a three-node high-availability system with incremental data replication and automatic failover, achieving approximately 2-minute recovery from primary server failures.

Department of CSIE, National Taiwan University
Jul 2025 - Aug 2026

Lead System Administrator, System Administration Team

- Led the system administration team responsible for maintaining and improving department-wide infrastructure, including network services, web services, DNS, computer labs, workstations, and printing systems.

NTU Student Cluster Competition Team
Jul - Nov 2025

HPC Student Cluster Engineer

- Built and optimized a 10 kW HPC cluster, achieving 561.6 TFLOP/s on HPL at SC25.
- Helped win 1st Place (Overall Winner) at the world's premier student supercomputing competition.

TEACHING EXPERIENCE

Department of Computer Science and Information Engineering, National Taiwan University

Advanced Network and System Administration
Spring 2026

Teaching Assistant

Network and System Administration
Spring & Fall 2025

Teaching Assistant (Spring); Lead Teaching Assistant (Fall)

Parallel Programming
Fall 2025

Teaching Assistant